

Mean-Variance Optimal Portfolios in the Presence of a Benchmark with Applications to Fraud Detection

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Abstract

We first study mean-variance efficient portfolios when there are no trading constraints and show that optimal strategies perform poorly in bear markets. We then assume that investors use a stochastic benchmark (linked to the market) as a reference portfolio. We derive mean-variance efficient portfolios when investors aim to achieve a given correlation (or a given dependence structure) with this benchmark. We also provide upper bounds on Sharpe ratios and show how these bounds can be useful for fraud detection. For example, it is shown that under some conditions it is not possible for investment funds to display a negative correlation with the financial market and to have a positive Sharpe ratio. All the results are illustrated in a Black-Scholes market.

Key-words: Mean-variance, Fraud detection, Optimal portfolio, Correlation constraints.

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1 Introduction

Markowitz (1952) and Roy (1952) were first in proposing a quantitative approach to determine the optimal trade-off between mean (return) and variance (risk). Their framework is nowadays known as mean-variance analysis and has become very influential as it combines algebraic simplicity with practical applicability. Markowitz pursued the study of optimal investment portfolios and his seminal works initiated a tremendous amount of research heading in several directions, ranging from the study of other notions for measuring risk, multi-period models (Mossin (1968), Cui et al. (2013)), non-negative final wealth (Korn and Trautmann (1995)) and imperfect markets (Xia and Yan (2006), Lim (2004)) to the inclusion of ambiguity on the returns (Goldfarb and Iyengar (2003)) or an uncertain horizon (Martellini and Urošević (2006)). See also Zhang et al. (2009) and Leung et al. (2012). More recently, several authors have been working on quadratic hedging or mean-variance hedging, which corresponds to the problem of approximating, with minimal mean squared error, a given payoff by the final value of a self-financing trading strategy in a financial market; see e.g., Lim (2006), Pham (2000) and Schweizer (1992), (2010), to cite only a few.

In an important contribution, Basak and Chabakauri (2010) have fully characterized time-consistent dynamic mean-variance optimal strategies. At any date prior to maturity, a time-consistent optimal strategy is the best possible mean-variance efficient allocation of wealth, assuming that an optimal mean variance strategy is also selected at each later instant in time. Mean-variance optimal strategies that are derived in a static setting¹ violate time-consistency in the sense that it may become optimal for a mean-variance investor to deviate away from this optimal mean-variance strategy during the investment horizon. However, these optimal strategies (derived in the static setting) can still be justified by assuming that the investor is pre-committed at time 0 and thus executes the dynamic investment strategy that has been decided at time $t = 0$. While time-consistency is a natural requirement, the assumption of pre-commitment is compatible with an investment practice in which a (retail) investor purchases a financial contract (from a financial institution) and does not trade (herself) afterwards.² It also fits with the behavior of an investment manager who revisits (optimizes) his portfolio periodically and sticks to his strategy between two dates. In practice, managers and other investors may also have additional constraints when optimizing their portfolio. One motivation for having constraints is that optimal (unconstrained) strategies are typically long with the market index and perform poorly in poor economic situations (Bernard et al. (2011)). In this paper, we show that the static setting is well suited to deal with a certain type of constraints that we motivate economically. See also Wang and Forsyth (2011) for a numerical approach of mean variance efficiency in a time-consistent framework and numerical comparisons of pre-committed strategies and time-consistent strategies.

Traditional mean-variance optimization consists in finding the best pre-committed allocation of assets assuming a buy-and-hold strategy or a constant-mix strategy (which requires a dynamic rebalancing to ensure a constant percentage invested in each asset). The ques-

¹By a “static setting”, we mean that the strategy is derived at the initial time $t = 0$ as the mean-variance efficient optimum with respect to the terminal wealth W_T without consideration for its properties at intermediate dates.

²This is also consistent with the work of Goldstein, Johnson and Sharpe (2008) who propose a tool that allows consumers to specify their desired probability distribution of terminal wealth at maturity.

tion raises then how pre-committed mean-variance efficient portfolios can be derived when all strategies are allowed and available. Of course, allowing for more trading strategies and thus more degrees of freedom will further enhance optimality. The *first contribution* in this paper is to derive optimal mean-variance strategies in this setting. We show that the optimal portfolio consists of a short position in the stochastic discount factor used for pricing derivatives and a long position in cash. We are also able to compute the maximum possible Sharpe ratio (Sharpe (1967)) of an optimal (mean-variance efficient) strategy. Bounds on the Sharpe ratio can be useful to regulators or other market participants for fraud detection, i.e., to assess whether the reported performance of a strategy is feasible or not. Recall for example that the Sharpe ratio of Madoff's strategy lied far above the maximum Sharpe ratio for plausible strategies (Bernard and Boyle (2009)).

In the second part of the paper we extend our study to the case when there is additional information on the strategy, for example on the way it interacts with the financial market or any other benchmark asset as a source of background risk. Our *second contribution* is then to derive tighter bounds on the Sharpe ratio. This is useful for improved fraud detection or abnormal performance reporting. For example, it is shown that under some conditions it is not possible for investment funds to display negative correlation with the financial market and to have a positive Sharpe ratio.

Considering the interaction with a benchmark asset is also a natural way to make mean-variance efficient strategies more resilient against declining markets. Indeed, the mean-variance efficient portfolios derived in the first part of the paper provide no protection against bear markets. In practice, many investors reward strategies that offer protection or, more generally, that exhibit some desired dependence with any other source of background risk (which we refer to as a benchmark). Our *third contribution* is to derive mean-variance optimal allocation schemes for investors who exhibit state-dependent preferences in the sense that they care about the first two moments of the strategy's distribution *and* additionally aim at obtaining a desired correlation or dependence with a benchmark asset.

The rest of the paper is organized as follows. The optimal portfolio problem and the assumptions on the financial market are presented in Section 2. Section 3 provides explicit expressions for mean-variance efficient portfolios when there are no trading constraints as well as a first application to fraud detection. Sections 4 and 5 extend these preliminary results to the case when there are constraints on the correlation (respectively the dependence) with a benchmark and illustrate how these results are particularly useful to improve fraud detection tools. Final remarks are presented in Section 6.

2 Market Setting

In this section, we provide our main assumptions and definitions. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space that describes a financial market. Assume that all market participants agree to use a (non-negative) stochastic discount process $(\xi_t)_t$ for pricing, i.e. the price at time 0 for a strategy with terminal payoff X_T (paid at time $T > 0$) writes as

$$c(X_T) = \mathbb{E}[\xi_T X_T]. \quad (1)$$

Note that the price of the unit cash-flow at time T is given by $c(1) = \mathbb{E}[\xi_T]$ and we define the risk-free rate r such that $e^{-rT} = \mathbb{E}[\xi_T]$. All payoffs X_T are assumed to be square integrable

ensuring that $c(X_T) < +\infty$. In particular $\text{var}(\xi_T) < +\infty$. We remark that this practice is usually motivated by assuming a frictionless and arbitrage-free financial market where the usual definition of absence of arbitrage is employed.³ In particular, we do not take into account transaction costs (Pelsser and Vorst (1996)). When the market is complete (all payoffs can be replicated) the stochastic discount factor ξ_T is uniquely given, but in general an infinite number of choices is possible. However, using a milder notion of arbitrage, Platen and Heath (2006) argue that under some conditions, the stochastic discount factor ξ_T corresponds to the inverse of the so-called Growth Optimal Portfolio (GOP)⁴ and also that the latter can be proxied by a market index. This motivates why in the remainder of the paper we refer to $1/\xi_t$ as “the market index” and we denote it by S_t^* . The pricing formula (1) can then be interpreted as the arithmetic average of the possible outcomes all expressed in units of the market index. Note how low values for the market index S_t^* correspond to high values for the discount factor ξ_t . This is consistent with economic theory in the sense that the states of a downturn are usually the most expensive states to insure and therefore correspond to the states ω where the highest values for the discount factor $\xi_T(\omega)$ are observed.

In the remainder of the paper, we consider an investor with a fixed horizon $T > 0$ without intermediate consumption. We denote by $W_0 > 0$ her initial wealth. For convenience, we assume that all ξ_t ($t > 0$) are continuously distributed.

3 Unconstrained Mean-Variance Optimal Portfolios

3.1 Mean-Variance Efficiency

Roy (1952) and Markowitz (1952) propose a quantitative approach to find mean-variance efficient allocation among risky assets assuming a buy-and-hold strategy. Their technique can also be applied in the context of constant-mix strategies. In this section we study mean-variance efficient portfolios when there are no restrictions on the possible strategies. Finding optimal policies turns out to be surprisingly simple. Indeed, let us first observe that an optimal mean-variance efficient final payoff X_T^* must necessarily be the cheapest possibility to generate a maximum mean for the given variance level.⁵ Otherwise it is easy to contradict the optimality of this payoff. Indeed, if the optimum is not the cheapest strategy then there is thus another strategy that is cheaper and also has maximum mean. The cost benefit can be invested in the risk free account and one obtains a strategy that has a higher mean for the same variance, which contradicts the mean-variance efficiency of the strategy.

Constructing a cheapest strategy amounts to minimizing the price (1). Observe that since $\text{std}(X_T^*)$ and $\text{std}(\xi_T)$ are both fixed and finite, minimizing the price (1) is equivalent

³The no-free-lunch with vanishing risk (NFLVR) concept is the prevalent way to describe absence of arbitrage. In their fundamental theorem of asset pricing, Delbaen and Schachermayer (1994) essentially state that NFLVR is equivalent to the existence of a stochastic discount factor (also called state-price density process).

⁴The Growth Optimal Portfolio is a diversified strategy which ultimately outperforms all other strategies with probability one. In the literature it also appears as the *Kelly portfolio*.

⁵In other words, the optimal mean-variance portfolio must be cost-efficient in the sense defined by Bernard et al. (2011).

to minimizing the correlation⁶ between X_T^* and the discount factor ξ_T . It is then a standard result in statistics that this occurs if and only if the optimal payoff X_T^* is linear in ξ_T (with a negative slope): the correlation is then minimized and equal to -1. The following proposition is now rather intuitive. A formal proof follows.

Proposition 3.1 (Mean-variance efficient portfolios). *Assume that the investor aims for a strategy that maximizes the expected return for a given variance s^2 for $s \geq 0$. The solution of the following mean-variance optimization problem*

$$\begin{cases} \max & \mathbb{E}[X_T] \\ \text{s.t.} & \begin{cases} \text{var}(X_T) = s^2 \\ c(X_T) = W_0 \end{cases} \end{cases} \quad (2)$$

is denoted by X_T^* and given as

$$X_T^* = a - b\xi_T,$$

where a and b are non-negative and equal to

$$a = (W_0 + b\mathbb{E}[\xi_T^2]) e^{rT}, \quad b = \frac{s}{\sqrt{\text{var}(\xi_T)}}. \quad (3)$$

Proof. Let us choose a and $b \geq 0$ such that $X_T^* = a - b\xi_T$ satisfies the constraints of Problem (2), that is a and b verify $\text{var}(X_T^*) = s^2$ and $c(X_T^*) = W_0$. We find (3). Observe that $\text{corr}(X_T^*, \xi_T) = -1$ and X_T^* is thus the unique payoff that is perfectly negatively correlated with ξ_T while satisfying the stated constraints. Consider next any other strategy X_T which also verifies these constraints (but is not negatively linear in ξ_T). We find that

$$\text{corr}(X_T, \xi_T) = \frac{\mathbb{E}[\xi_T X_T] - \mathbb{E}[\xi_T]\mathbb{E}[X_T]}{\sqrt{\text{var}(\xi_T)}\sqrt{\text{var}(X_T)}} > -1 = \text{corr}(X_T^*, \xi_T). \quad (4)$$

Since $\text{var}(X_T) = s^2 = \text{var}(X_T^*)$ and $\mathbb{E}[\xi_T X_T] = W_0 = \mathbb{E}[\xi_T X_T^*]$ it follows from (4) that

$$\mathbb{E}[\xi_T]\mathbb{E}[X_T] < \mathbb{E}[\xi_T]\mathbb{E}[X_T^*],$$

which shows that X_T^* maximizes the expectation and thus solves Problem (2). $\square$

Note that the maximal expected return $\mathbb{E}[X_T^*]$ is given as

$$\mathbb{E}[X_T^*] = (W_0 + b\text{var}(\xi_T)) e^{rT} \geq W_0 e^{rT}, \quad (5)$$

because $b \geq 0$, reflecting that taking risk results in a positive risk premium. Proposition 3.1 shows that, in case all trading is permitted, mean-variance efficient portfolios amount to holding a cash amount ($a \geq 0$) combined with a short position ($b \geq 0$) in the stochastic discount factor ξ_T . While the benefits are limited to receiving a at the maximum, the investor suffers unlimited losses when the market index $S_T^* = 1/\xi_T$ goes to zero. The monotonicity of the optimal payoff with the market index may not suit investors who may wish to have some degree of protection for their portfolio in bad economic scenarios (i.e. in a declining market). Adding constraints on correlation or on the dependence can partly address this issue as we will see later in Section 4.

⁶ $\text{corr}(X_T, \xi_T) = \frac{\mathbb{E}[\xi_T X_T] - \mathbb{E}[\xi_T]\mathbb{E}[X_T]}{\text{std}(\xi_T)\text{std}(X_T)}$. As the moments of ξ_T are given, it follows that for given moments $\mathbb{E}[X_T]$ and $\text{std}(X_T)$ minimizing $\mathbb{E}[\xi_T X_T]$ is equivalent to minimizing correlation $\text{corr}(X_T, \xi_T)$.

Rather than maximizing the expected return for a given (risk) variance, one can also consider the dual problem of minimizing the variance for a given expected return. We obtain the following result.

Proposition 3.2 (Mean-variance efficient portfolios - Dual setting). *Assume that the investor aims for a strategy that minimizes the variance for a given expected wealth. The solution of the following mean-variance optimization problem*

$$\min_{\begin{cases} \mathbb{E}[X_T] = W_0 e^{mT} \\ c(X_T) = W_0 \end{cases}} \text{var}(X_T) \quad (6)$$

where $m \geq r$, is denoted by X_T^* and given as

$$X_T^* = a - b\xi_T,$$

where a and b are non-negative and equal to $a = W_0 e^{mT} + b e^{-rT}$, $b = \frac{W_0(e^{(m-r)T} - 1)}{\text{var}(\xi_T)}$.

Proof. As the proof of Proposition 3.2 is rather similar to Proposition 3.1, we give it in Appendix A.1. $\square$

Note that the requirement $m \geq r$ is natural to ensure the problem is well-posed for mean-variance investors who prefer more to less. Otherwise, investing the initial wealth at the fixed risk-free rate r would always ensure higher final wealth with zero variance.

Mean-variance optimization is similar to maximizing quadratic utility. As the following result readily follows from Propositions 3.1 and 3.2, we omit its proof.

Corollary 3.1 (Optimizing quadratic utility). *Assume that the investor has a utility function $U(x) = x - \frac{\alpha}{2}x^2$ (where $0 < \alpha \leq \frac{e^{-rT}}{W_0}$ and $x \leq \frac{1}{\alpha}$)⁷. The solution of the following expected utility maximization problem*

$$\max_{c(X_T)=W_0} \mathbb{E}[u(X_T)] \quad (7)$$

is denoted by X_T^* and is given by $X_T^* = a - b\xi_T$, where a and b are non-negative and given by $a = \frac{1}{\alpha}$ and $b = \frac{e^{-rT} - \alpha W_0}{\alpha \mathbb{E}[\xi_T^2]}$.

3.2 Example in the Black-Scholes Setting

Let us show how the results apply in a Black-Scholes market. While this setting does not always allow to accurately reflect true market behavior, its tractability makes it a reference model used in practice and a traditional work-horse employed in the finance literature to develop new ideas and get insights. In the Black-Scholes setting, there is a bank account

⁷Note that imposing $\alpha \leq \frac{e^{-rT}}{W_0}$ is natural to ensure the investor is also considering (utility of) wealth levels $x > W_0 e^{rT}$, when optimizing her expected utility. In the opposite case, optimizing quadratic utility gives rise to portfolios that exhibit variability for a lower expected return than what can be obtained by simply investing at the fixed rate r . In addition, the requirement $x \leq \frac{1}{\alpha}$ is needed to ensure the utility function is non-decreasing on admissible wealth levels.

earning a constant risk-free rate $r > 0$ and a risky stock with the following dynamics under the real-world probability measure

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t. \quad (8)$$

Here W_t is a standard Brownian motion, $\mu > r$ is the instantaneous expected return and σ is the volatility. The unique stochastic discount factor process $(\xi_t)_t$ in the Black-Scholes setting is given by

$$\xi_t = e^{-rt} e^{-\theta W_t - \frac{1}{2}\theta^2 t}, \quad \theta = \frac{\mu - r}{\sigma}. \quad (9)$$

Furthermore, we also find that $\frac{dS_t^*}{S_t^*} = \left(\frac{\theta}{\sigma}\mu + (1 - \frac{\theta}{\sigma})r\right) dt + \theta dW_t$ so that the market index S_t^* amounts to a constant mix strategy, where at time t a fraction $\frac{\theta}{\sigma}$ is invested in the risky asset and the remaining fraction $1 - \frac{\theta}{\sigma}$ in the bank account. The solution to the *mean-variance* optimization problem (2) is thus given by

$$X_T^* = a - \frac{b}{S_T^*}, \quad (10)$$

where a, b are non-negative and given by $a = W_0 e^{rT} + s \frac{e^{\theta^2 T}}{\sqrt{e^{\theta^2 T} - 1}}$ and $b = \frac{se^{rT}}{\sqrt{e^{\theta^2 T} - 1}}$. It is clear that a similar result can be derived for Problem (6).

3.3 Maximum Sharpe Ratio and Application to Fraud Detection

The Sharpe ratio is a well-known measure balancing risk (variance) and return (mean) of a portfolio X_T (see Sharpe (1967)). It is defined as

$$SR(X_T) = \frac{\mathbb{E}[X_T] - W_0 e^{rT}}{\text{std}(X_T)}.$$

It is clear that mean-variance optimality of a portfolio is tied to the maximality of the Sharpe ratio.

Proposition 3.3 (Maximal Sharpe Ratio). *All mean-variance efficient portfolios X_T^* have the same maximal Sharpe ratio given by*

$$SR^* = \sqrt{\frac{\text{var}(\xi_T)}{\mathbb{E}^2[\xi_T]}} = e^{rT} \text{std}(\xi_T). \quad (11)$$

Furthermore, a portfolio is mean-variance efficient if and only if it has maximal Sharpe ratio.

Proof. The proof of this proposition is given in Appendix A.2. □

In the specific context of a Black-Scholes market (see (8)), Proposition 3.3 can also be found in Goetzmann et al. (2002)⁸ but our result is more general and holds in a fairly

⁸In the appendix in Goetzmann et al. (2002) the maximal Sharpe ratio is proved to be $\exp(\theta^2 \Delta t) - 1$, where in the context of this paper Δt is the investment horizon. Note that this result does not appear in the published version of Goetzmann et al. (2006).

general market setting. This maximum Sharpe ratio applies for all mean-variance efficient strategies as well as any admissible trading strategy in the financial market. Hence it can be used for fraud detection in the sense that it allows to trace Sharpe ratio of fund managers which are too high to be “feasible”. When the Sharpe ratio of a strategy violates the upper bound SR^* , this can be interpreted as a signal that there might be a fraud, or a mistake regarding the reported returns. In practice, to apply fraud detection one needs to construct an estimator for the Sharpe ratio which can be achieved using the reported returns of the fund at hand. The upper bound itself requires the estimation of the standard deviation of the stochastic discount factor ξ_T . While this variable is not really observable in the market, several methods exist to estimate its distributional properties; see for example Aït-Sahalia and Lo (2001). We now pursue the basic idea of fraud detection in some more detail in the Black-Scholes setting.

Fraud detection in the Black-Scholes model: In the Black-Scholes market one finds from (9) that $\mathbb{E}[\xi_T] = e^{-rT}$ and $\mathbb{E}[\xi_T^2] = e^{-2rT}e^{\theta^2T}$. Hence, in this setting the expression (11) for the maximal Sharpe-ratio SR^* can be made explicit and we find that

$$SR^* = \sqrt{e^{\theta^2T} - 1}. \quad (12)$$

Using observed market data one can now readily estimate SR^* and next compare Sharpe ratios of funds (derived from reported data) with this maximum. This was used by Bernard and Boyle (2009) to show that the returns from the option strategy pursued by Madoff were too good to be true.

Fraud detection in a general market: The Black-Scholes setting does not always comply with real markets which implies that fraud detection based on the expression (12) is prone to some model error. A possible way to address this drawback consists in a better use of the available market data allowing to obtain a non-parametric estimator for SR^* . The following proposition exhibits the Sharpe ratio in terms of observable option prices.

Proposition 3.4 (Fraud detection). *Assume that $\xi_T = f(S_T)$ (where f is typically decreasing and S_T is the risky asset) and that all European call options on the underlying S_T maturing at $T > 0$ are traded.⁹ Let $C(K)$ denote the price of a call option on S_T with strike K . Then, the Sharpe ratio $SR(X_T)$ of any admissible strategy with payoff X_T satisfies the following upper bound*

$$SR(X_T) \leq \sqrt{e^{2rT} \int_0^{+\infty} f(K) \frac{\partial^2 C(K)}{\partial K^2} dK - 1}. \quad (13)$$

Proof. The proof of Proposition 3.4 is given in Appendix A.3. □

Note that in an incomplete market, the pricing kernel ξ_T is not unique anymore. The upper bounds (11) and (13) are then still valid upper bounds for the Sharpe ratio of a given strategy X_T (with respect to the pricing kernel ξ_T that is used to price this strategy).

⁹There is often a large (but finite) number of options available in the market, in which case assuming a continuum of strikes could be seen as a reasonable approximation of reality. Carr and Chou (1997) note that it is “analogous to the continuous trading assumption permeating the continuous time literature”. See also Breeden and Litzenberger (1978).

However, these bounds may not be attainable anymore because the market is incomplete. In other words, a dynamic strategy that achieves the maximum Sharpe ratio may not exist (if $a - b\xi_T$ is not attainable). The fact that $a - b\xi_T$ is not attainable is not a real problem for the application to fraud detection since it is still an upper bound.

It is important to understand that the violation of the upper bound as derived in Proposition 3.4, or as given in (12) for the specific case of a Black-Scholes model, has to be seen as an indication (a signal) that there could be a fraud, but not as a formal proof of it. We already explained that the calculation of the upper bound is subject to model error, thus a violation of this upper bound does not always imply a fraud. There might be other reasons that explain why an observed manager's Sharpe ratio can be higher than the upper bound even if there is no fraud.

Firstly, the Sharpe ratio of a strategy is based on the average and the standard deviation. In practice, one has to estimate these two moments and hence only confidence intervals for the Sharpe ratio can be obtained but not the true value. Hence, the observed Sharpe ratio may violate the upper bound but this fact is not necessarily (statistically) meaningful.¹⁰ Secondly, the upper bound that we propose has been derived in a continuous time framework by optimizing over all self-financing strategies that are adapted to the market information on the security prices. In practice, the manager may be able to capture information outside the financial market and to use this extra information to optimize his portfolio. In particular, a manager with insider information has a strategy that is also (partially) based on this "future information" and thus not adapted to the prices filtration. In this case, the Sharpe ratio of such strategy may lie above the theoretical bound (that is derived among all adapted self-financing strategies). A violation of the upper bound may thus also serve as a potential signal for detecting insider trading. Similarly, a manager who is rebalancing his portfolio on a daily basis may use an extra source of information to decide if he goes long or short in the risky asset for the next business day. There is then always a chance that he consistently makes the right decision and thus achieves a higher Sharpe ratio (by luck). In the context of a multidimensional financial market with n stocks, there is a small probability to always pick the best stocks while rebalancing the portfolio. Being an outstanding stock picker was one of the arguments used by Madoff to explain the exceptionally high Sharpe ratio of his investment strategy.

To conclude, it is important to understand that the violation of the upper bound derived in Proposition 3.4 or given in (12) for the specific case of a Black-Scholes model should be seen as a signal that there could be a fraud, but not as a proof.

4 Mean-Variance Efficiency with a Correlation Constraint

The fraud detection mechanism described in the previous section can be greatly improved by taking into account additional information available in the market. Indeed, the maximum Sharpe ratios derived in Proposition 3.3 and 3.4 do not take into account the dependence

¹⁰It is well-documented in the literature that optimal portfolio choice is subject to parameter uncertainty, in that a small perturbation of the inputs may lead to a large change in the optimal portfolio. Robust portfolio selection techniques have been developed to deal with this issue (Ben-Tal and Nemirovski (1998), Goldfarb and Iyengar (2003), Tütüncü and Koenig (2004)). For a complete discussion of robust portfolio optimization and the associated solution methods, see Fabozzi et al. (2007) and the references therein.

features between the investment strategy and the financial market. Regulators, however, can estimate the Sharpe ratio of a hedge fund but can also investigate correlations of the fund with indices in the market, and this additional source of information may be useful for refining the process of fraud detection. A popular strategy amongst hedge funds is the so-called “market-neutral” strategy. One of its key properties is that it typically ensures very low correlation with market indices. We show that in this instance the maximum possible Sharpe ratio is significantly reduced compared to the unconstrained case.

Furthermore, recall that unconstrained mean-variance efficient payoffs have bounded gains but no protection against a market crash giving rise to unlimited losses (Propositions 3.1 and 3.2). An investor may then be interested in choosing the dependence with a benchmark of her choice so that, for instance, her investment strategy is no longer decreasing when the market index decreases. It appears as a suitable reference point for her investment, and it allows to better control the states in which cash-flows are received. The worst outcomes for the strategy do no longer necessarily coincide with bad scenarios for the market index.

Hence, in this section we focus on investors who care about the first two moments of the distribution of a strategy *and* additionally aim at obtaining a desired correlation (Section 4) or dependence (Section 5) with a benchmark asset B_T .

4.1 Mean-Variance Efficiency

The following proposition gives a mean-variance optimal allocation policy for an investor with a fixed correlation constraint.

Proposition 4.1 (Optimal portfolio with a correlation constraint). *Let B_T be a benchmark which is linearly independent from ξ_T with $\text{var}(B_T) < +\infty$. Let $|\rho| < 1$ and $s > 0$. Consider the following mean-variance optimization problem*

$$\begin{aligned} & \max \mathbb{E}[X_T] \\ & \left\{ \begin{array}{l} \text{var}(X_T) = s^2 \\ c(X_T) = W_0, \\ \text{corr}(X_T, B_T) = \rho \end{array} \right. \end{aligned} \quad (14)$$

Let a , b and c be uniquely determined by the set of equations

$$\begin{aligned} \rho &= \text{corr}(cB_T - \xi_T, B_T) \\ s &= b\sqrt{\text{var}(\xi_T - cB_T)} \\ W_0 &= ae^{-rT} - b(E[\xi_T^2] - cE[\xi_T B_T]), \end{aligned}$$

then X_T^* given as $X_T^* = a - b(\xi_T - cB_T)$, is a solution to Problem (14).

Proof. Consider the function $f(c) := \text{corr}(cB_T - \xi_T, B_T)$ and observe that $\lim_{c \rightarrow -\infty} f(c) = -1$ and $\lim_{c \rightarrow +\infty} f(c) = 1$. Since $f(c)$ is continuous it follows from the intermediate value theorem that the equation $\rho = f(c)$ has a solution when solving for c . Moreover, this solution is unique as

$$f'(c) = \frac{\text{var}(B_T)\text{var}(\xi_T) - (\text{cov}(\xi_T, B_T))^2}{(\text{std}(cB_T - \xi_T))^3 \text{std}(B_T)} > 0.$$

It is now clear that $X_T^* = a - b(\xi_T - cB_T)$ is the unique payoff that is linear in $\xi_T - cB_T$ while satisfying all constraints and note that $b > 0$. It remains to show that it is the optimal solution to Problem (14). Hence, consider any other payoff X_T that satisfies the constraints and which is non-linear in $\xi_T - cB_T$. We have that

$$\text{corr}(X_T, \xi_T - cB_T) = \frac{\mathbb{E}[X_T(\xi_T - cB_T)] - \mathbb{E}[\xi_T - cB_T]\mathbb{E}[X_T]}{\text{std}(\xi_T - cB_T)\text{std}(X_T)} > -1 = \text{corr}(X_T^*, \xi_T - cB_T)$$

Since both X_T and X_T^* satisfy the constraints we have that $\text{std}(X_T) = \text{std}(X_T^*)$, $\mathbb{E}[X_T\xi_T] = \mathbb{E}[X_T^*\xi_T]$ and $\text{cov}(X_T, B_T) = \text{cov}(X_T^*, B_T)$. Hence the inequality holds true if and only if $\mathbb{E}[X_T^*] > \mathbb{E}[X_T]$. This ends the proof. $\square$

Remark 4.1. In the Problems (2), (6) and (14), the equalities in the constraints can sometimes be replaced by inequalities without impacting the solution. First, the solution to Problem (2) is also the portfolio, which maximizes the expectation such that the variance is less than s^2 instead of being equal to s^2 (as we do). This feature clearly appears from the solution $a - b\xi_T$, as b decreases if s decreases and thus the expected final value of the portfolio decreases as well. Similarly, in the dual problem (6), the solution that minimizes the variance of the terminal value of the portfolio for a given expected value is also the solution to a variance minimization with minimum expected value. However, for the correlation constraint in Problem (14), replacing the equality by an inequality usually affects the solution. For example, let us assume that we replace the constraint “ $\text{corr}(X_T, B_T) = \rho$ ” by the constraint “ $\text{corr}(X_T, B_T) \leq \rho$ ” and also assume that the optimum $a - b\xi_T$ that was derived in the unconstrained case (as a solution to Problem (2)) satisfies this inequality constraint. Clearly, the unconstrained optimum $a - b\xi_T$ (for Problem (2)) will then also be the optimum for the constrained Problem (14). However, it is clear that one can also have that $\text{corr}(a - b\xi_T, B_T) > \rho$ and in this instance the unconstrained solution fails to be the optimum of the constrained Problem (14). Note also that it is possible to solve Problem (14) in presence of inequality constraints on the correlation, i.e. when $\rho_1 \leq \text{corr}(X_T, B_T) \leq \rho_2$ for some $-1 \leq \rho_1 \leq \rho_2 \leq 1$. Finding the optimal strategy consists of two-step optimization in which we derive first the optimum, say X_T^ρ , of Problem (14) for each $\rho_1 \leq \rho \leq \rho_2$ and next we find the optimal solution by optimizing $\mathbb{E}[X_T^\rho]$ on the interval $[\rho_1, \rho_2]$.

Note that in the case when the benchmark B_T is not a function of the stochastic discount factor ξ_T (or equivalently of the market index S_T^*), the worst case outcomes for the optimal strategy $X_T^* = a - b(\xi_T - cB_T)$ do not necessarily occur when the market index is low. Bad outcomes for the market index S_T^* might be compensated by good outcomes for the benchmark asset. In contrast, when B_T and thus also X_T^* depends on S_T^* only, then the optimum may or may not provide protection against downturns (depending on its precise functional relationship with the market index). We now provide an illustration of the theoretical results.

Illustration in the Black-Scholes Market when the benchmark $B_T = S_T^*$: Consider a Black-Scholes market as in (8). Let us solve Problem (14) when the benchmark B_T is the market index, that is $B_T = S_T^*$. From Proposition 4.1, the optimal solution is of the form $X_T^* = a - b(\xi_T - cS_T^*)$, where c is computed from the equation $\rho = \text{corr}(cS_T^* - \xi_T, S_T^*)$, b is derived from $b = \frac{s}{\sqrt{\text{var}(\xi_T - cS_T^*)}}$ and $a = W_0 e^{rT} + b(e^{-2rT + \theta^2 T} - c)e^{rT}$.

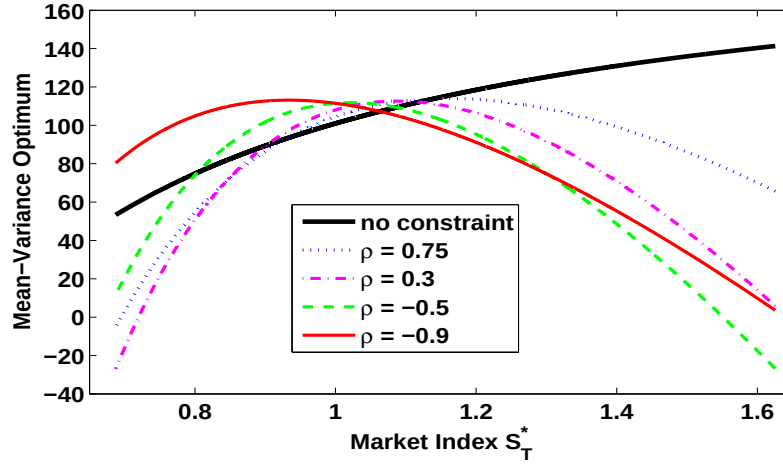


Figure 1: Optimal payoffs as a function of the market index S_T^* , for different values of the correlation ρ with the benchmark S_T^* using the following parameters: $W_0 = 100$, $r = 0.05$, $\mu = 0.07$, $\sigma = 0.2$, $T = 1$, $S_0 = 100$, $s = 10$.

Figure 1 shows that constraining the strategy to be (sufficiently) negatively correlated can improve the returns during a crisis (when S_T^* is low). However this goes at the cost of a lower performance when the market index increases significantly. Note that the unconstrained optimal strategy also appears as an optimal strategy in presence of a (redundant) correlation constraint, that is when $\rho = \text{corr}(a - b\xi_T, S_T^*)$ (which is close to 1 in this example).

4.2 Maximum Sharpe Ratio and Application to Fraud Detection

It is clear that adding constraints reduces degrees of freedom for trading strategies and thus lowers the maximum possible Sharpe ratio. We obtain the following proposition.

Proposition 4.2 (Constrained Maximal Sharpe Ratio). *Any mean-variance efficient portfolio X_T^* which satisfies the additional constraint $\text{corr}(X_T^*, B_T) = \rho$ with a benchmark asset B_T (that is not linearly dependent to ξ_T) has the same maximum Sharpe ratio SR_ρ^* given by*

$$SR_\rho^* = e^{rT} \frac{\text{cov}(\xi_T, \xi_T - cB_T)}{\text{std}(\xi_T - cB_T)} \leq SR^* = e^{rT} \text{std}(\xi_T). \quad (15)$$

where SR^* is the unconstrained Sharpe ratio found in Proposition 3.3 and where c is determined uniquely by the equation $\text{corr}(\xi_T - cB_T, B_T) = \rho$.

Proof. The proof of Proposition 4.2 is given in Appendix A.4. □

Note that the constrained strategy exhibits a strictly lower Sharpe ratio unless $c = 0$ which just means that the unconstrained optimal strategy happens to satisfy the correlation constraint. The proposition shows that for fraud detection it is useful to incorporate correlation features of displayed returns. We illustrate this point in a numerical example in

the next paragraph in the Black-Scholes setting. In particular, we observe that maximum Sharpe ratios can be significantly lower than in the unconstrained case.

Illustration in the Black-Scholes Market: Figure 2 displays the maximum Sharpe ratio for the unconstrained case (which is approximately equal to 0.1 for the parameter set used in Figure 2) and constrained Sharpe ratio for different levels of correlation constraints when the market index S_T^* is the benchmark.

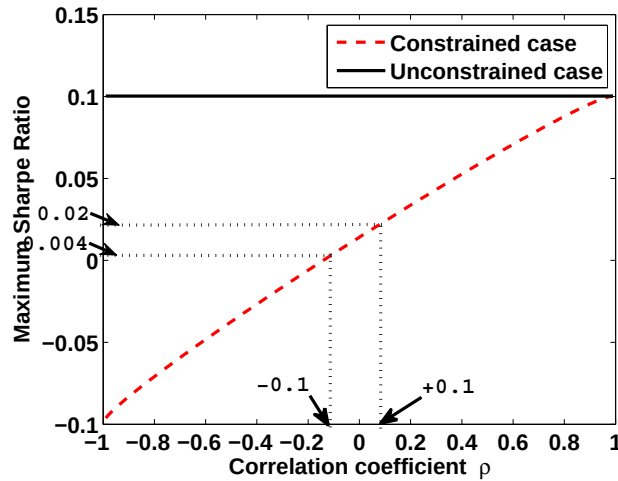


Figure 2: Maximum Sharpe ratio SR_ρ^* given by (15) for different values of the correlation ρ when the benchmark is $B_T = S_T^*$. We use the following parameters: $W_0 = 100$, $r = 0.05$, $\mu = 0.07$, $\sigma = 0.2$, $T = 1$, $S_0 = 100$.

Observe that for low correlation levels ($\rho \in [-0.1, 0.1]$), the maximum Sharpe ratio is 0.02 only, which is five times lower than when there are no constraints. Adding the information on the dependence between the strategy's performance and the market index can thus greatly improve fraud detection. Observe also that for negative correlation levels, the maximum Sharpe ratio can be negative¹¹, thus if hedge fund returns display a negative correlation with the financial market and a positive Sharpe ratio, then there could be some suspicion about these returns. This is strongly different from what we observed in the unconstrained case as the maximum Sharpe ratio is always positive in this case because of (5).

Finally, observe that the constrained case reduces to the unconstrained one when ρ is equal to the correlation of the unconstrained optimum $a - b\xi_T$ with the market index S_T^* (which happens when ρ is close to 1). The constraint is thus redundant in that case.

¹¹At first, it might look counter intuitive that an optimal strategy has a lower return than what can be achieved risk-free. However, enforcing a negative dependence with the market comes at some cost. A similar observation can be drawn for the put option: it has a low expected return but provides protection when markets fall.

4.3 Mean-Variance Optimal Portfolios with Multiple Correlation Constraints

In practice, investors may consider more than one benchmark when making investment decisions. Likewise, a fund manager's portfolio returns are observed in conjunction with the returns of many other market indices. In particular, one may have at hand a correlation matrix between a given fund and n market indices, and having this information allows to develop an improved upper bound on the Sharpe ratio. In this section, we derive optimal strategies (and thus also the maximum Sharpe ratio) amongst strategies that satisfy the correlation matrix. Using not only information about the marginal distribution (computed as the Sharpe ratio of the portfolio) but also the correlation matrix significantly improves the upper bound, and thus may facilitate fraud detection.

To do so, consider investors who care about the first and second moments of the distribution of their investment strategy but now in addition aim at fixing the correlations with n linearly independent benchmark assets B_T^i ($i = 1, 2, \dots, n$). Their correlation matrix Σ (i.e. $\Sigma_{ij} := \text{corr}(B_T^i, B_T^j)$) is positive definite. Without loss of generality, we can further assume that all benchmarks have expectation 0 and variance 1.

It is well-known that there is a unique Cholesky decomposition $\Sigma = AA^\top$ where A is a triangular matrix. It also follows that the vector $\mathbf{D}^\top = (D_T^1, D_T^2, \dots, D_T^n)$ determined through $\mathbf{D} = A^{-1}\mathbf{B}$ with $\mathbf{B}^\top = (B_T^1, B_T^2, \dots, B_T^n)$ is a vector of orthogonal benchmarks all with expectation 0 and variance 1. It is now clear that the following *mean-variance* optimization problem

$$\begin{cases} \max & \mathbb{E}[X_T] \\ \text{var}(X_T) = s^2 \\ c(X_T) = W_0, \\ \text{corr}(X_T, B_T^i) = \rho_i \quad (i = 1, 2, \dots, n) \end{cases} \quad (16)$$

can also be alternatively formulated as

$$\begin{cases} \max & \mathbb{E}[X_T] \\ \text{var}(X_T) = s^2 \\ c(X_T) = W_0, \\ \text{cov}(X_T, D_T^i) = d_i \quad (i = 1, 2, \dots, n) \end{cases} \quad (17)$$

for appropriate choices for d_i .

Proposition 4.3 (Optimal portfolio with multiple correlation constraints). *The optimal solution X_T^* to Problem (17) when the d_i satisfy¹² $\sum_{i=1}^n d_i^2 \leq s^2$ (with $s^2 > 0$) is given by*

$$X_T^* = a - b\xi_T + \sum_{i=1}^n c_i D_T^i, \quad (18)$$

¹²This condition is needed for the problem to be well posed. Note indeed that X_T^* can be expressed as $X_T^* = E[X_T^*] + \sum_{i=1}^n \text{cov}(\xi_T, D_T^i) D_T^i + \varepsilon$ where ε is a random variable with zero mean that is orthogonal (uncorrelated) to the different D_T^i ($i = 1, 2, \dots, n$).

where a , b and c_i ($i = 1, 2, \dots, n$) are as follows:

$$\begin{aligned} c_i &= d_i + b \operatorname{cov}(\xi_T, D_T^i) \\ b &= \sqrt{\frac{s^2 - \sum_{i=1}^n d_i^2}{\operatorname{var}(\xi_T - \sum_{i=1}^n \operatorname{cov}(\xi_T, D_T^i) D_T^i)}} \\ a &= W_0 e^{rT} + b e^{rT} \mathbb{E}[\xi_T^2] + \sum_{i=1}^n c_i e^{rT} c(D_T^i). \end{aligned}$$

Proof. The proof is given in Appendix A.5. □

5 Mean-Variance Optimal Portfolios with a Dependence Constraint

Correlation is only one property related to dependence. It measures the linear relationship between strategies but falls short in depicting dependence fully. A useful device for reflecting the interaction between the strategy's payoff X_T and B_T is the copula. Indeed Sklar's theorem shows that the joint distribution of (B_T, X_T) can be decomposed as

$$\mathbb{P}(B_T \leq y, X_T \leq x) = C(F_{B_T}(y), F_{X_T}(x)), \quad (19)$$

where C is the joint distribution (also called the copula) for a pair of uniform random variables U and V over $(0, 1)$ and where F_{B_T} and F_{X_T} denote respectively the (marginal) cdf of B_T and X_T . Hence, the copula $C : [0, 1]^2 \rightarrow \mathbb{R}$ fully describes the interaction between the strategy's payoff X_T and the benchmark B_T . A constraint on the copula is much more informative than a correlation constraint. A copula is a function and contains full information about the interaction between two variables. A correlation is a single number and cannot describe the complex nature of dependence fully.¹³ Similarly, Value-at-Risk is only a single number and will never describe the risk as well as knowing the exact loss distribution.

The partial derivative $c_u(v) := \frac{\partial}{\partial u} C(u, v)$ has an interesting property, namely it can be interpreted as a conditional probability:

$$c_u(v) = \mathbb{P}(V \leq v | U = u). \quad (20)$$

Property (20) is extremely useful for constructing payoffs with desired dependence properties. For example, $c_U(V)$ is a uniform variable that depends on U and V and which is independent of U . Conversely, if U and T are independent uniform random variables then $c_U^{-1}(T)$ is a uniform variable (depending on U and T) that has copula C with U .

The following propositions give mean-variance optimal allocation schemes in the presence of a benchmark. We use the notation $C(\cdot, \cdot)$ to reflect the desired dependence for the couple (B_T, X_T^*) that is between the benchmark B_T and the final value of the investment strategy X_T^* at the investment horizon T .

¹³Note that even when two variables are normally distributed their dependence is not described by their correlation coefficient. Knowledge of the correlation coefficient is sufficient for depicting the dependence when the variables follow a bivariate normal distribution, but this is an assumption which actually imposes a lot of structure on the interaction between both variables.

5.1 Market Index as Benchmark

The first proposition in this section provides mean-variance efficient portfolios when the market index S_T^* , and thus ξ_T , is used as the benchmark.

Proposition 5.1 (Optimal portfolio when the market index is the benchmark). *Let $B_T = \xi_T$. For $t \in (0, T)$, define the variable $\mathcal{A}_t$ as*

$$\mathcal{A}_t = \left(c_{F_{\xi_T}(\xi_T)} \right)^{-1} \left[j_{F_{\xi_T}(\xi_T)}(F_{\xi_t}(\xi_t)) \right],$$

where the functions $j_u(v)$ and $c_u(v)$ are defined as the first partial derivative for $(u, v) \rightarrow J(u, v)$ and $(u, v) \rightarrow C(u, v)$ respectively, and where J denotes the copula for the random pair (ξ_T, ξ_t) . Assume that $\mathbb{E}[\xi_T | \mathcal{A}_t]$ is strictly decreasing in $\mathcal{A}_t$. For $s > 0$, a solution of the following constrained mean-variance optimization problem

$$\begin{cases} \max & \mathbb{E}[X_T] \\ \text{s.t.} & \text{var}(X_T) = s^2 \\ & c(X_T) = W_0 \\ & C : \text{copula between } X_T \text{ and } B_T \end{cases} \quad (21)$$

is given by X_T^* ,

$$X_T^* = a - b \mathbb{E}[\xi_T | \mathcal{A}_t]. \quad (22)$$

Here a, b are non-negative and given by $a = (W_0 + b \mathbb{E}[\xi_T \mathbb{E}[\xi_T | \mathcal{A}_t]]) e^{rT}$, $b = \frac{s}{\text{std}(\mathbb{E}[\xi_T | \mathcal{A}_t])}$.

Note that the maximal expected return of X_T^* is given as

$$\begin{aligned} \mathbb{E}[X_T^*] &= (W_0 + b \text{var}(\mathbb{E}[\xi_T | \mathcal{A}_t])) e^{rT} \\ &> W_0 e^{rT}. \end{aligned} \quad (23)$$

Actually, comparing (23) with (5), one observes that adding dependence constraints decreases the expected return of the optimal strategy and this decrease is directly proportional to $\text{var}(\xi_T) - \text{var}(\mathbb{E}[\xi_T | \mathcal{A}_t])$.

Proof. The proof of Proposition 5.1 can be found in Appendix A.6 □

We now formulate a few important remarks.

Remark 5.1. (i) In general, the Mean-Variance optimal portfolios discussed in this section are not unique as the choice of t is arbitrary. (ii) One may wonder what happens if in Proposition 5.1 the non-decreasingness property for $\mathbb{E}[\xi_T | \mathcal{A}_t]$ is not fulfilled. Then, it still follows that mean-variance efficient portfolios, provided they exist, must write as $f(\mathcal{A}_t)$ for some non-decreasing f . Unfortunately, it is then not clear how to find a function f that minimizes $\text{corr}(f(\mathcal{A}_t), \xi_T)$, or equivalently, that minimizes $\text{corr}(f(\mathcal{A}_t), \mathbb{E}[\xi_T | \mathcal{A}_t])$. (iii) When $\mathbb{E}[\xi_T | \mathcal{A}_t]$ is increasing in $\mathcal{A}_t$ then it holds for the mean-variance efficient portfolio $f(\mathcal{A}_t)$ (provided it exists) that $\mathbb{E}[f(\mathcal{A}_t)] \leq W_0 e^{rT}$, meaning that the imposed benchmark constraint comes at significant cost.

5.2 General Benchmark

We remark from the previous proposition that fixing the dependence with ξ_T does not generally result in unique mean-variance efficient portfolios. However, when the benchmark is not functionally dependent with ξ_T , the optimal allocation can become unique as the following proposition shows.

Proposition 5.2 (Constrained Mean-Variance Efficiency). *Let $s > 0$. Assume that the benchmark B_T has a joint density with ξ_T . Define the variable $\mathcal{A}$ as*

$$\mathcal{A} = \left(c_{F_{B_T}(B_T)} \right)^{-1} \left[j_{F_{B_T}(B_T)}(1 - F_{\xi_T}(\xi_T)) \right], \quad (24)$$

where the functions $j_u(v)$ and $c_u(v)$ are defined as the first partial derivative for $(u, v) \rightarrow J(u, v)$ and $(u, v) \rightarrow C(u, v)$ respectively, and where J denotes the copula for the random pair (B_T, ξ_T) . Furthermore, assume that $\mathbb{E}[\xi_T | \mathcal{A}]$ is decreasing in $\mathcal{A}$. Then, the solution to the problem

$$\begin{cases} \max & \mathbb{E}[X_T] \\ \text{s.t.} & \text{var}(X_T) = s^2 \\ & c(X_T) = W_0 \\ & C : \text{copula between } X_T \text{ and } B_T \end{cases} \quad (25)$$

is uniquely given as

$$X_T^* = a - b\mathbb{E}[\xi_T | \mathcal{A}],$$

where a, b are non-negative and equal to $a = (W_0 + b\mathbb{E}[\xi_T \mathbb{E}[\xi_T | \mathcal{A}]]) e^{rT}$ and $b = \frac{s}{\text{std}(\mathbb{E}[\xi_T | \mathcal{A}])}$.

Similarly as the case when the benchmark is the market index, the maximum expected return verifies

$$\mathbb{E}[X_T^*] = (W_0 + b\text{var}(\mathbb{E}[\xi_T | \mathcal{A}])) e^{rT} \geq W_0 e^{rT}.$$

5.3 Application to the Black-Scholes Market

We specialize the results by considering a Gaussian dependence¹⁴ and by taking the market index as the benchmark.

Proposition 5.3 (Case $B_T = S_T^*$). *Let $B_T = S_T^*$. Assume that $\rho_0 \geq 0$. Then, the solution to the problem (21) when the copula C is the Gaussian copula with correlation ρ_0 , i.e. $C_{\rho_0}^{\text{Gauss}}$, is given by X_T^* ,*

$$X_T^* = a - bG_T^c. \quad (26)$$

Here G_T is a weighted average of the benchmark and the market index. It is given as $G_T = (S_t^*)^\alpha S_T^*$ with α ,

$$\frac{1}{\alpha} = \frac{t}{T} \left(\rho_0 \sqrt{\frac{T-t}{t} \frac{1}{1-\rho_0^2}} - 1 \right),$$

¹⁴We say that two variables X and Y have a gaussian dependence with correlation coefficient ρ_0 if (X, Y) is distributed as $(f(N), g(M))$, where f and g are increasing, and where N and M are (bivariate) standard normally distributed random variables that exhibit a correlation coefficient equal to ρ_0 .

and where the coefficients a , b and c are given as

$$a = W_0 e^{rT} + b e^{rT} \mathbb{E}[\xi_T G_T^c], \quad b = \frac{s}{\sqrt{\text{var}(G_T^c)}}, \quad c = -\frac{\alpha t + T}{(\alpha + 1)^2 t + (T - t)}.$$

Proof. The proof of Proposition 5.3 can be found in Appendix A.8. $\square$

Proposition 5.4 (Case $B_T = S_t^*$). *Let $B_T = S_t^*$ ($0 < t < T$) be the benchmark. Assume that $\rho_0 \geq -\sqrt{1 - \frac{t}{T}}$. Then, the solution to Problem (25) when the copula C is the Gaussian copula with correlation ρ_0 , $C_{\rho_0}^{\text{Gauss}}$ is given by X_T^* ,*

$$X_T^* = a - b G_T^c. \quad (27)$$

Here G_T is a weighted average of the benchmark and the market index. It is given as $G_T = (S_t^*)^\alpha S_T^*$ with α ,

$$\alpha = \rho_0 \sqrt{\frac{T-t}{t} \frac{1}{1-\rho_0^2}} - 1,$$

where the coefficients a , b and c are given as

$$a = W_0 e^{rT} + b e^{rT} \mathbb{E}[\xi_T G_T^c], \quad b = \frac{s}{\sqrt{\text{var}(G_T^c)}}, \quad c = -\frac{\alpha t + T}{(\alpha + 1)^2 t + (T - t)}.$$

Proof. The proof of Proposition 5.4 can be found in Appendix A.9. $\square$

The Black-Scholes setting allows to derive an explicit expression for the Maximal Sharpe ratio. We obtain the following result.

Proposition 5.5 (Constrained Maximal Sharpe Ratio). *All mean-variance efficient portfolios X_T^* which satisfy the additional constraint that the copula between their final value X_T^* and the market portfolio S_t^* (for all $t \leq T$) is the Gaussian copula, have the same maximal Sharpe ratio $SR_{\rho_0, G}^*$ given by*

$$SR_{\rho_0, G}^* = \frac{e^{rT} \mathbb{E}[\xi_T G_T^c] - \mathbb{E}[G_T^c]}{\text{std}(G_T^c)} \leq SR^*. \quad (28)$$

Here $\mathbb{E}[G_T^c] = e^{M + \frac{V}{2}}$ and $\text{var}(G_T^c) = (e^V - 1)e^{2M + V}$, with $M := \mathbb{E}[\ln(G_T^c)] = c(r + \frac{\theta^2}{2})(\alpha t + T)$ and $V := \text{var}(\ln(G_T^c)) = c^2 \theta^2 (\alpha^2 t + T + 2\alpha t)$. Moreover $\mathbb{E}[\ln(\xi_T) + \ln(G_T^c)] = M - rT - \frac{\theta^2}{2}T$ and $\text{var}(\ln(\xi_T) + \ln(G_T^c)) = \theta^2 (c^2 \alpha^2 t + (c - 1)^2 T + 2c(c - 1)\alpha t)$ so that $\mathbb{E}[\xi_T G_T^c]$ reflects the expectation of a lognormal which can be computed from these two first log-moments similarly as we did for G_T^c .

Proof. We omit the proof as it is a straightforward calculation. $\square$

Remark 5.2. Proposition 5.5 is fundamentally different from the result in Proposition 4.2, which only derived the maximum Sharpe ratio with a given correlation. Imposing that the dependence between the portfolio and a benchmark is specified by a Gaussian copula with given correlation coefficient is more informative than imposing a correlation constraint as done in Proposition 4.2. There are indeed infinitely many copulas that can result in the same level of correlation and the Gaussian copula is just one example. In practice it is

easier to estimate the correlation than the copula and Proposition 4.2 might thus be more useful for practical application. However, if one has a good idea of the desired copula, the result in Proposition 5.5 will be strictly better in the sense that the upper bound will be smaller and thus the fraud detection scheme will be improved.

Observe that the constrained case reduces to the unconstrained maximum Sharpe ratio when the correlation in the Gaussian copula is $\rho_0 = \sqrt{t/T}$. The reason is that the copula between the unconstrained optimum and S_t^* is the Gaussian copula with correlation $\rho_0 = \sqrt{t/T}$. The constraint is thus redundant in that case. Let us illustrate this last proposition by a numerical example in the Black-Scholes setting.

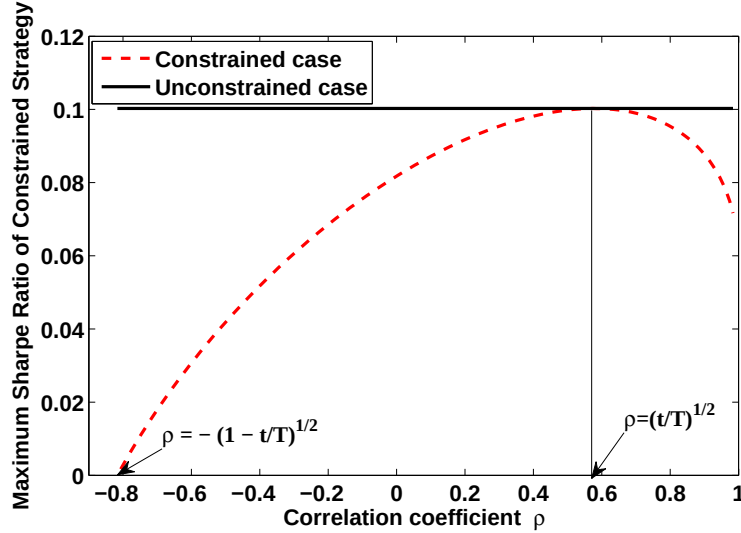


Figure 3: Maximum Sharpe ratio $SR_{\rho_0, G}^*$ given by (28) for different values of the correlation ρ_0 when the benchmark is $B_T = S_t^*$. We use the following parameters: $t = 1/3$, $\sqrt{t/T} = 0.577$, $-\sqrt{1-t/T} = -0.816$, $W_0 = 100$, $r = 0.05$, $\mu = 0.07$, $\sigma = 0.2$, $T = 1$, $S_0 = 100$.

6 Final Remarks

In this paper we first analyze mean-variance optimal portfolios when there are no constraints on trading and dependence. We show that optimal portfolios consist in having a short position in the stochastic discount factor. Next, we depart from the classical setting and assume the investor also seeks for portfolios satisfying dependence constraints. We are able to provide optimal portfolios and also derive bounds for maximum possible Sharpe ratios. Throughout the paper we explore how the results can be useful for fraud detection.

The approach that we propose for dealing with state-dependent constraints is technically rooted in the theory of stochastic dependence and bounds on copulas. In a recent paper, Bernard et al. (2013) characterize optimal strategies for investors who specify dependence with a benchmark, under worst-case scenarios only, and show that the new strategies outperform traditional diversified strategies; see also Bernard et al. (2012) for the technical

background. Finally, Bernard & Vanduffel (2013) explore the connections between the pricing of financial and insurance (non-financial) claims and bounds on copulae are a technical device in doing so.

Throughout the paper we dealt with several examples of state-dependent constraints by looking at the dependence between the portfolio and the benchmark. The static setting considered in the paper allows to solve for explicit optimal strategies in the presence of these constraints. It is not straightforward to handle such constraints in a time-consistent setting (Basak and Chabakauri (2010)) and we leave it for future research. Another interesting future direction is to improve optimal mean-variance hedging using our results.

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A Proofs

A.1 Proof of Proposition 3.2

Let us choose a and $b \geq 0$ such that $X_T^* = a - b\xi_T$ satisfies the constraints of Problem (6), that is where a and b verify $\mathbb{E}(X_T^*) = W_0 e^{mT}$ and $c(X_T^*) = W_0$. We find that

$$a = W_0 e^{mT} + b e^{-rT}, \quad b = \frac{W_0 (e^{(m-r)T} - 1)}{\text{var}(\xi_T)}.$$

Note that $\text{corr}(X_T^*, \xi_T) = -1$ and thus also that $\text{cov}(X_T^*, \xi_T) < 0$. Let us show that X_T^* solves the optimization problem (6). Consider any other strategy X_T that verifies the constraints of Problem (6). As there is a unique linear strategy negatively correlated with ξ_T that satisfies the constraints, $\text{corr}(X_T, \xi_T) > -1$. Observe that

$$\text{corr}(X_T, \xi_T) = \frac{\mathbb{E}[\xi_T X_T] - \mathbb{E}[\xi_T] \mathbb{E}[X_T]}{\sqrt{\text{var}(\xi_T)} \sqrt{\text{var}(X_T)}} > -1 = \text{corr}(X_T^*, \xi_T) \quad (29)$$

As X_T satisfies the constraints of Problem (6), we have that $\mathbb{E}[X_T] = W_0 e^{mT} = \mathbb{E}[X_T^*]$ and $\mathbb{E}[\xi_T X_T] = W_0 = \mathbb{E}[\xi_T X_T^*]$ so that $\text{cov}(X_T, \xi_T) = \text{cov}(X_T^*, \xi_T) < 0$. Therefore (29) implies that

$$\sqrt{\text{var}(X_T)} > \sqrt{\text{var}(X_T^*)}$$

which shows that X_T^* minimizes the variance and solves Problem (6). Note that it is unique in doing so because it is the only payoff with correlation -1 that satisfies the constraints of the optimization problem. $\square$

A.2 Proof of Proposition 3.3

Note first that a strategy that maximizes the Sharpe ratio is clearly a solution to Problems (2) and (6). Consider then a mean-variance efficient portfolio X_T^* solution to either (2) or (6). In both cases it is straightforward to show that

$$SR^* := SR(X_T^*) = \frac{a - b e^{-rT} - W_0 e^{rT}}{b \sqrt{\text{var}(\xi_T)}} = e^{rT} \sqrt{\text{var}(\xi_T)}.$$

Hence, all mean-variance efficient portfolio X_T^* have the same Sharpe-ratio SR^* . $\square$

A.3 Proof of Proposition 3.4

Define the risk-neutral probability measure Q as

$$e^{-rT} \mathbb{E}_Q[X_T] = \mathbb{E}[\xi_T X_T].$$

Apply Proposition 3.3 and observe that $\sqrt{\text{var}(\xi_T)} = \sqrt{e^{-rT} \mathbb{E}_Q[\xi_T] - e^{-2rT}}$ because $\mathbb{E}[\xi_T^2] = e^{-rT} \mathbb{E}_Q[\xi_T]$ and $\mathbb{E}[\xi_T] = e^{-rT}$. Furthermore $\mathbb{E}_Q[\xi_T] = \mathbb{E}_Q[f(S_T)]$. It is also well-known that the density of S_T under the risk-neutral probability Q can be obtained as $g(K) = e^{rT} \frac{\partial^2 C(K)}{\partial K^2}$.

Then $\sqrt{\text{var}(\xi_T)} = \sqrt{\int_0^{+\infty} f(K) \frac{\partial^2 C(K)}{\partial K^2} dK - e^{-2rT}}$. $\square$

A.4 Proof of Proposition 4.2

Similarly as the proof of Proposition 3.3, it is clear that a portfolio with maximum Sharpe ratio, maximizes expectation with a given variance. Therefore its final payoff is of the form $X_T^* = a - b\xi_T + bcB_T$ (see Proposition 4.1). It satisfies the budget constraint $\mathbb{E}[\xi_T X_T^*] = W_0$ and the correlation constraint with the benchmark B_T , $\text{corr}(X_T^*, B_T) = \rho_0$. Therefore

$$a = W_0 e^{rT} + b e^{rT} \mathbb{E}[\xi_T^2] - b c e^{rT} \mathbb{E}[\xi_T B_T].$$

The maximum Sharpe ratio can be computed as

$$SR_\rho^* = \frac{\mathbb{E}[X_T^*] - W_0 e^{rT}}{\text{std}(X_T^*)}.$$

After some simplification, using the expression for a , and simplifying by b , one obtains the first part of (15):

$$SR_\rho^* = e^{rT} \frac{\text{cov}(\xi_T, \xi_T - cB_T)}{\text{std}(\xi_T - cB_T)}$$

Note that this expression depends on c but c is uniquely determined by the correlation constraint $\text{corr}(X_T^*, B_T) = \rho$.

Finally it is clear that SR_ρ^* is smaller than the unconstrained Sharpe ratio SR^* as

$$SR_\rho^* = e^{rT} \frac{\text{cov}(\xi_T, \xi_T - cB_T)}{\text{std}(\xi_T - cB_T)} \leq e^{rT} \frac{\text{std}(\xi_T) \text{std}(\xi_T - cB_T)}{\text{std}(\xi_T - cB_T)} = SR^*$$

and the inequality is strict as B_T is non-linear in ξ_T . This ends the proof. $\square$

A.5 Proof of Proposition 4.3

Proof. (i) We first show that the stated X_T^* verifies the constraints. To this end, we rewrite X_T^* as

$$X_T^* = a - b \left(\xi_T - \sum_{i=1}^n \text{cov}(\xi_T, D_T^i) D_T^i \right) + \sum_{i=1}^n d_i D_T^i,$$

and it is easily verified that $\text{cov}(X_T^*, D_T^i) = d_i$ for all $i = 1, 2, \dots, n$. Furthermore, taking into account that $\xi_T - \sum_{i=1}^n \text{cov}(\xi_T, D_T^i) D_T^i$ is orthogonal to each D_T^j , it follows that $\text{var}(X_T^*) = s^2$. Finally, it is straightforward that $c(X_T^*) = W_0$. (ii) It remains to show that X_T^* is the (unique) solution to Problem (14). Hence, consider any other payoff X_T satisfying the constraints. Then we have that

$$\text{corr} \left(X_T, b\xi_T - \sum_{i=1}^n c_i D_T^i \right) > -1 = \text{corr} \left(X_T^*, b\xi_T - \sum_{i=1}^n c_i D_T^i \right)$$

Taking into account that both X_T and X_T^* satisfy the constraints we find that this inequality holds true if and only if

$$-\mathbb{E}[X_T]\mathbb{E}\left[\sum_{i=1}^n c_i D_T^i\right] - \mathbb{E}\left[b\xi_T - \sum_{i=1}^n c_i D_T^i\right] \mathbb{E}[X_T] > -\mathbb{E}[X_T^*]\mathbb{E}\left[\sum_{i=1}^n c_i D_T^i\right] - \mathbb{E}\left[b\xi_T - \sum_{i=1}^n c_i D_T^i\right] \mathbb{E}[X_T^*].$$

and thus, since $b > 0$, if and only if $\mathbb{E}[X_T^*] > \mathbb{E}[X_T]$. $\square$

A.6 Proof of Proposition 5.1

Recall that a copula is invariant by increasing transformation. An optimal solution X_T , whenever it exists, writes as $f(U)$ where f is increasing and where U is a uniform random variable having copula C with ξ_T . We observe that taking U as $\mathcal{A}_t$ provides us with a uniform distributed random variable with the right dependence property and it remains to determine f . To this end we choose a and $b \geq 0$ such that $X_T^* = a - b\mathbb{E}[\xi_T|\mathcal{A}_t]$ satisfies the constraints of Problem (21), that is a and b verify $\text{var}(X_T^*) = s^2$ and $c(X_T^*) = W_0$. We find that

$$\begin{aligned} a &= (W_0 + b\mathbb{E}[\xi_T\mathbb{E}(\xi_T|\mathcal{A}_t)]) e^{rT}, \\ b &= \frac{s}{\sqrt{\text{var}(\mathbb{E}[\xi_T|\mathcal{A}_t])}}. \end{aligned}$$

Since $\mathbb{E}[\xi_T|\mathcal{A}_t]$ is strictly decreasing in $\mathcal{A}_t$, X_T^* has the right dependence with $B_T = \xi_T$. It is also clear that $\text{corr}(X_T^*, \mathbb{E}[\xi_T|\mathcal{A}_t]) = -1$ and X_T^* is the unique payoff that is perfectly negatively correlated with $\mathbb{E}[\xi_T|\mathcal{A}_t]$ and also satisfies all the constraints of Problem (21). Consider next any other strategy X_T which also verifies these constraints. We find that

$$\text{corr}(X_T, \mathbb{E}[\xi_T|\mathcal{A}_t]) = \frac{\mathbb{E}[\mathbb{E}[\xi_T|\mathcal{A}_t]X_T] - \mathbb{E}[\xi_T]\mathbb{E}[X_T]}{\sqrt{\text{var}(\mathbb{E}[\xi_T|\mathcal{A}_t])}\sqrt{\text{var}(X_T)}} > -1 = \text{corr}(X_T^*, \mathbb{E}[\xi_T|\mathcal{A}_t]). \quad (30)$$

Since X_T satisfies the constraints of Problem (21), we have that $\text{var}(X_T) = s^2 = \text{var}(X_T^*)$ and $\mathbb{E}[\xi_T X_T] = \mathbb{E}[\mathbb{E}[\xi_T|\mathcal{A}_t]X_T] = W_0 = \mathbb{E}[\xi_T X_T^*]$. Therefore (30) implies that

$$\mathbb{E}[\xi_T]\mathbb{E}[X_T] < \mathbb{E}[\xi_T]\mathbb{E}[X_T^*],$$

which shows that X_T^* maximizes the expectation and thus is a candidate for solving Problem (2). Assume that there is another variable $\mathcal{B}_t$ that is uniformly distributed and having copula C with ξ_T , and let Y_T^* be a possible solution that is a function of $\mathcal{B}_t$. Since $(\xi_T, \mathcal{B}_t)$ and $(\xi_T, \mathcal{A}_t)$ have the same joint distribution it is easy to see that another potential optimal strategy Y_T^* satisfying the constraints must also write as $Y_T^* = a - b\mathbb{E}[\xi_T|\mathcal{B}_t]$. Since Y_T^* has the same distribution than X_T^* it follows that both X_T^* and Y_T^* must have the same expectation and thus X_T^* is an optimal solution. $\square$

A.7 Proof of Proposition 5.2

Consider a payoff X_T and assume it is optimal. From property (20) it follows that U defined as,

$$U = j_{F_{B_T}(B_T)}(1 - F_{\xi_T}(\xi_T)),$$

is the (almost surely) unique uniformly distributed random variable that is independent of B_T and conditionally (on B_T) decreasing in ξ_T . Consider now the payoff $F_{X_T|B_T}^{-1}(U)$ and note that $(B_T, F_{X_T|B_T}^{-1}(U))$ and (B_T, X_T) have the same joint distribution. Hence, the payoff $F_{X_T|B_T}^{-1}(U)$ also satisfies the stated variance and copula constraint. It also has the same expectation as X_T . However, $F_{X_T|B_T}^{-1}(U)$ is always less expensive than X_T because

$$\begin{aligned} c(X_T) = \mathbb{E}[X_T \xi_T] &= \mathbb{E}[\mathbb{E}[X_T \xi_T | B_T]] \geq \mathbb{E}\left[\mathbb{E}\left[F_{X_T|B_T}^{-1}(U) \xi_T \mid B_T\right]\right] \\ &= \mathbb{E}\left[F_{X_T|B_T}^{-1}(U) \xi_T\right] = c\left(F_{X_T|B_T}^{-1}(U)\right) \end{aligned}$$

where the inequality follows from the fact that $F_{X_T|B_T}^{-1}(U)$ and ξ_T are conditionally (on B_T) anti-monotonic, and equality can only occur when $X_T = F_{X_T|B_T}^{-1}(U)$ almost surely (this is a key property needed in the proofs of Bernard et al. (2011), see appendix, proof of property (2) and proposition 1 in Bernard et al. (2011)). If this is not the case then there exists $\alpha > 0$ so that the new payoff $F_{X_T|B_T}^{-1}(U) + \alpha$ satisfies all constraints while having a strictly higher expectation than X_T . Hence, we conclude that optimal payoffs must be of the form $F_{X_T|B_T}^{-1}(U)$ or equivalently of the form $F_{X_T}^{-1}(\mathcal{A})$ for some non-decreasing function $F_{X_T}^{-1}$ which remains to be determined. Consider $F_{X_T}^{-1}(x) = a - b\mathbb{E}[\xi_T | \mathcal{A} = x]$ where a and $b \geq 0$ are such that $X_T^* = a - b\mathbb{E}[\xi_T | \mathcal{A}]$ satisfies the constraints of Problem (25). Note indeed that $\mathbb{E}[\xi_T | \mathcal{A}]$ is assumed to be decreasing so that we only need to ensure that a and b verify $\text{var}(X_T^*) = s^2$ and $c(X_T^*) = W_0$. We find,

$$a = (W_0 + b\mathbb{E}[\xi_T \mathbb{E}[\xi_T | \mathcal{A}]]) e^{rT}, \quad b = \frac{s}{\text{std}(\mathbb{E}[\xi_T | \mathcal{A}])}.$$

It now holds that $\text{corr}(X_T^*, \mathbb{E}[\xi_T | \mathcal{A}]) = -1$. Let us show that X_T^* solves Problem (25). Consider any other payoff $X_T = g(\mathcal{A})$, non-linear in $\mathbb{E}[\xi_T | \mathcal{A}]$ satisfying the constraints and note that $\text{corr}(X_T, \mathbb{E}[\xi_T | \mathcal{A}]) > -1$ holds. Observe that

$$\text{corr}(X_T, \mathbb{E}[\xi_T | \mathcal{A}]) = \frac{\mathbb{E}[\mathbb{E}[\xi_T | \mathcal{A}] X_T] - \mathbb{E}[\xi_T] \mathbb{E}[X_T]}{\sqrt{\text{var}(\mathbb{E}[\xi_T | \mathcal{A}])} \sqrt{\text{var}(X_T)}} > -1 = \text{corr}(X_T^*, \mathbb{E}[\xi_T | \mathcal{A}]).$$

Since X_T satisfies the constraints of Problem (25), we have that we have that $\text{var}(X_T) = s^2 = \text{var}(X_T^*)$ and $\mathbb{E}[\xi_T X_T] = \mathbb{E}[\mathbb{E}[\xi_T | \mathcal{A}] X_T] = W_0 = \mathbb{E}[\xi_T X_T^*]$. Therefore it follows that

$$\mathbb{E}[\xi_T] \mathbb{E}[X_T] < \mathbb{E}[\xi_T] \mathbb{E}[X_T^*],$$

which shows that X_T^* maximizes the expectation and is the unique solution to Problem (25). $\square$

A.8 Proof of Proposition 5.3

Let us notice that $F_{\xi_t}(\xi_t) = 1 - F_{S_t^*}(S_t^*)$. Then, similar as in the proof of Proposition 5.4 on finds that,

$$j_{F_{S_T^*}(S_T^*)}(F_{\xi_t}(\xi_t)) = \Phi \left(\frac{\ln \left(\frac{S_t^*}{(S_T^*)^{\frac{t}{T}}} \right)}{\vartheta \sqrt{\frac{t(T-t)}{T}}} \right).$$

and also that

$$\begin{aligned} \mathcal{A}_t &:= c_{F_{S_T^*}(S_T^*)}^{-1} \left(j_{F_{S_T^*}(S_T^*)}(F_{\xi_t}(\xi_t)) \right) \\ &= \Phi \left[\sqrt{1 - \rho_0^2} \left(\frac{\ln \left(\frac{S_t^* S_0^{\frac{t}{T}-1}}{(S_T^*)^{\frac{t}{T}}} \right)}{\vartheta \sqrt{\frac{t}{T}(T-t)}} \right) + \rho_0 \left(\frac{\ln \left(\frac{S_T^*}{S_0^*} \right) - \left(\mu^* - \frac{\sigma^2}{2} \right) T}{\theta \sqrt{T}} \right) \right]. \end{aligned}$$

Furthermore, let $\frac{1}{\alpha} = \frac{t}{T} \left(\rho_0 \sqrt{\frac{T-t}{t} \frac{1}{1-\rho_0^2}} - 1 \right)$, and note that

$$E \left[\frac{1}{S_T^*} \middle| \mathcal{A}_t \right] = E \left[\frac{1}{S_T^*} \middle| \alpha W_t^* + W_T^* \right] = \gamma G_T^\delta,$$

where $\gamma \geq 0$ is the appropriate coefficient. $G_T = (S_t^*)^\alpha S_T^*$ and

$$\delta = \text{cov}(-W_T^*, \alpha W_t^* + W_T^*) = -\frac{\alpha t + T}{(\alpha + 1)^2 + (T - t)}.$$

Note that

$$\delta < 0 \iff \alpha > -\frac{T}{t} \iff \rho_0 > 0.$$

Hence, applying Proposition 5.1 we find that the optimal strategy writes as $X_T^* = a - bG_T^c$, where a and $b > 0$ follow from the budget and variance constraint respectively. We find,

$$a = W_0 e^{rT} + b e^{rT} \mathbb{E}[\xi_T G_T^c], \quad b = \frac{s}{\sqrt{\text{var}(G_T^c)}}. \quad \square$$

A.9 Proof of Proposition 5.4

We first notice that due to the inverse monotonic relation between ξ_t and S_t^* it holds for all t that $F_{\xi_t}(\xi_t) = 1 - F_{S_t^*}(S_t^*)$. An easy calculation shows that,

$$F_{S_t^*}(S_t^*) = \Phi \left(\frac{\ln(S_t^*) - \left(\mu^* - \frac{\sigma^2}{2} \right) t}{\theta \sqrt{t}} \right),$$

where $\mu^* = \frac{\theta}{\sigma}\mu + (1 - \frac{\theta}{\sigma})r$. Furthermore for any Gaussian copula C one has

$$c_u(v) := \frac{\partial C}{\partial u}(u, v) = \Phi \left[\frac{\Phi^{-1}[v] - \rho_0 \Phi^{-1}[u]}{\sqrt{1 - \rho_0^2}} \right],$$

so that

$$c_u^{-1}(w) = \Phi \left[\sqrt{1 - \rho_0^2} \Phi^{-1}[w] + \rho_0 \Phi^{-1}[u] \right].$$

Hence, we find after some calculations that

$$j_{F_{S_t^*}(S_t^*)}(1 - F_{\xi_T}(\xi_T)) = j_{F_{S_t^*}(S_t^*)}(F_{S_T^*}(S_T^*)) = \Phi \left(\frac{\ln \left(\frac{S_T^*}{S_t^*} \right) - (\mu^* - \frac{\theta^2}{2})(T - t)}{\theta \sqrt{T - t}} \right)$$

and also

$$\begin{aligned} \mathcal{A} &:= c_{F_{S_t^*}(S_t^*)}^{-1} \left(j_{F_{S_t^*}(S_t^*)}(F_{S_T^*}(S_T^*)) \right) \\ &= \Phi \left[\sqrt{1 - \rho_0^2} \left(\frac{\ln \left(\frac{S_T^*}{S_t^*} \right) - (\mu^* - \frac{\theta^2}{2})(T - t)}{\theta \sqrt{T - t}} \right) + \rho_0 \left(\frac{\ln(S_t^*) - (\mu^* - \frac{\theta^2}{2})t}{\theta \sqrt{t}} \right) \right]. \end{aligned}$$

Let $\alpha = \rho_0 \sqrt{\frac{T-t}{t} \frac{1}{1-\rho_0^2}} - 1$ and observe that that $\mathcal{A}$ is monotonic in $\alpha \theta W_t^* + \theta W_T^*$. Hence,

$$E \left[\frac{1}{S_T^*} \mid \mathcal{A} \right] = E \left[\frac{1}{S_T^*} \mid \alpha \theta W_t^* + \theta W_T^* \right]$$

Then, we can re-express $E \left[\frac{1}{S_T^*} \mid \mathcal{A} \right]$ as $E \left[\frac{1}{S_T^*} \mid \mathcal{A} \right] = \gamma G_T^c$. Here, $G_T = (S_t^*)^\alpha S_T^*$, $\gamma \geq 0$ is the appropriate coefficient and the parameter c writes as

$$c = \frac{\text{cov}(-\vartheta W_T^*, \alpha \vartheta W_t^* + \vartheta W_T^*)}{\text{var}(\alpha \vartheta W_t^* + \vartheta W_T^*)} = -\frac{\alpha t + T}{(\alpha + 1)^2 t + (T - t)}.$$

Clearly, $E \left[\frac{1}{S_T^*} \mid \mathcal{A} \right]$ is decreasing in $\mathcal{A}$ if and only if $c < 0$. One finds that

$$c < 0 \iff \alpha > -\frac{T}{t} \iff \rho_0 > -\sqrt{1 - \frac{t}{T}}.$$

Hence, applying Proposition 5.2 one has that the optimal strategy writes as $X_T^* = a - b G_T^c$, where a and $b > 0$ follow from the budget and variance constraint respectively. We find,

$$a = W_0 e^{rT} + b e^{rT} \mathbb{E}[\xi_T G_T^c], \quad b = \frac{s}{\sqrt{\text{var}(G_T^c)}} \quad \square$$